# Routing

## 1. Static Vs Dynamic routing Protocol

| Basis for comparision | Static routing | Dynamic routing |
|---|---|---|
| Configuration | Manual | Automatic |
| Routing table building | Routing locations are hand-typed. | Locations are dynamicaly filled in the table |
| Routes | User defined | Routes are updated according to change in topology. |
| Routing algorithms | Doesn't employ complex algorithm. | Uses complex algorithm to perform routing operations. |
| Implemented in | Small network | Large network |
| Link failure | obstruct the rerouting | doesn't affect the rerouting |
| Security | Provides high security | Less secure due to broadcasts & multicasts |
| Routing Protocal | No routing protocal needed | Routing protocal such as RIP, EIGP etc are involved. |

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## 2. Intradomain Vs interdomain routing

| Intradomain routing | Interdomain routing |
|---|---|
| 1. Routing algorithm works only within domains. | 1. Routing algorithm works within & between domains. |
| 2. It need to know only about their other routers within their domain. | 2. It need to know only about other routers within and between their domain. |
| 3. Protocals used in intradomain routing are known as Interior-gateway Protocols. | 3. Protocols used in interdomain routing are known as Exterior-gateway Protocols. |
| 4. Routing takes place within an autonomous network. | 4. Routing takes place between the autonomous network. |
| 5. Some Popular Protocols of this routing are RIP & OSPF. | 5. Popular Protocols of this routing is BGP used to connect two or more autonomous system. |

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3. Distance vector Vs Link state routing Protocol (17)

| Basis for comp. | Distance vector | Link state |
|---|---|---|
| Algorithm | Bellman ford | Dijsktra |
| Network view | Topology information from the neighbour point of view | Complete information on the network topology. |
| Best Path calc. | Based on the least number of hops. | Based on the cost |
| Updates | Full routing table | Link state updates |
| Updates frequency | Periodic updates | Triggered updates |
| CPU & memory | Low utilisation | Intensive |
| Simplicity | High | Requires a trained network administrator |
| Convergence time | Moderate | Fast |
| Updates | on broadcast | on multicast |
| Hierarchical structure | No | Yes |
| Intermidiate nodes | No | Yes |

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4. Discuss two major classes of routing Protocol. (19)

1. Distance vector routing Protocol :

These Protocols selects the best path in the basis of hop counts to reach a destination network in the particular direction. Dynamic Protocol like RIP is an example of distance vector routing protocol. Hop counts is each router which occurs in between the source and the destination network. The path with the least hop count will be choosen as the best path.

Features :

- updates of network are exchanged periodically.
- updates is always broadcast.
- Full routing tables are sent in updates.
- Routers always trust on routing information recieved from neighbour routers. This is also known as routing on rumors.

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## 2. Link state routing Protocol:

These Protocols know more about internetwork than any other distance vector routing Protocol. These are also known as SPF (Shortest Path first) Protocol. OSPF is an example of link state routing protocol.

### Features:

- Hello messages, also known as keep alive messages are used for neighbour discovery and recovery.
- Concept of triggered updates are used ie updates are triggered only when there is a topology change.
- only that much updates are exchanged which is requested by the neighbour routers.

Maintain 3 table —

| | |
|---|---|
| Neighbor table | The table which contains the information about the neighbors of the router only. |
| Topology table | This table contains the whole information about the whole topology. |
| Routing table | This tables contains all the best routes to the advertised network. |

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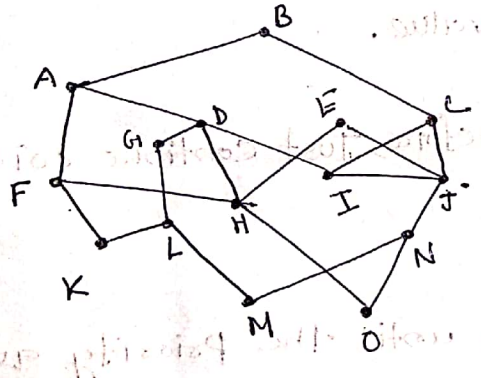
## 5. Short notes :

### (i) Default route :

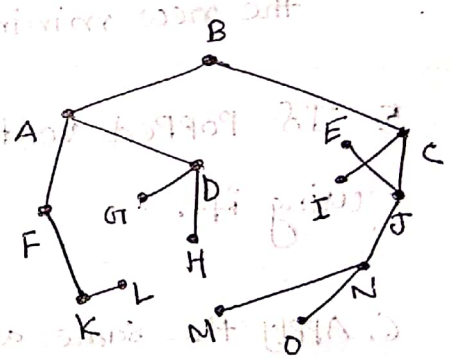
- A default route is a network route used by a router when no other known route exists for a given IP packet's destination address.
- All the packets for destination which is not known by the router's routing table are sent to the default route.

### (ii) Sink tree : (19,17)

- The set of optimal routes to a particular node forms a sink tree.
- Sink ~~trees~~ trees are n't necessarily unique
- Goal of all routing algorithms is to discover sink trees for all destinations




A-Subnet




Sink tree rooted at B

### (iii) Flooding :

- Flooding is used in computer networks routing algorithm in which every incoming packet is sent through every outgoing link except the one it arrived on.
- Flooding is used in bridging
- Super simple to implement, but generates a lot of redundant packets

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## 6. Dijkstra algorithm:

**Algorithm:**

1. Set all vertices distance = infinity except for the source vertex, set the source distance = 0. (dist[S] = 0, dist[V] = ∞)
2. Push the source vertex in a min-Priority queue.
3. Pop the vertex with the minimum distance from the Priority queue (at first poped vertex = source)
4. For each new adjacent node u,

   If dist[v] + weight (u,v) < dist[u] then there is a minimal distance found for u, so update the dist[u] to the new minimal distance value.
5. If popped vertex is visited before, just continue without using it.
6. Apply the same algorithm again until the Priority queue is empty.

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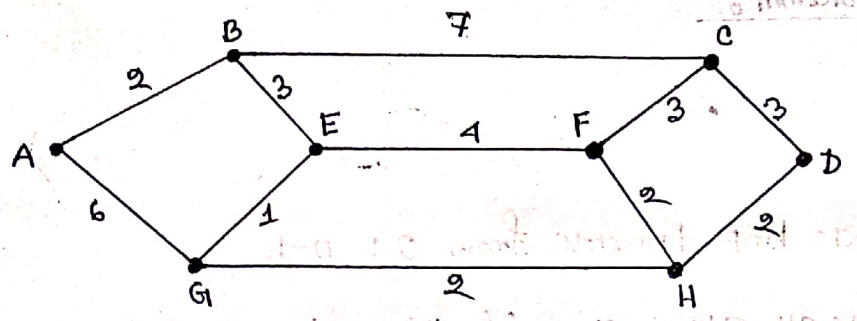
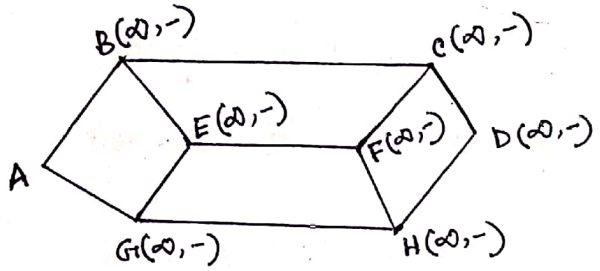
(19,17)

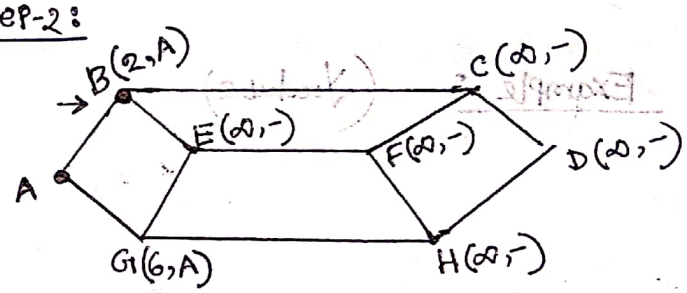



Find the shortest Path from A to D?

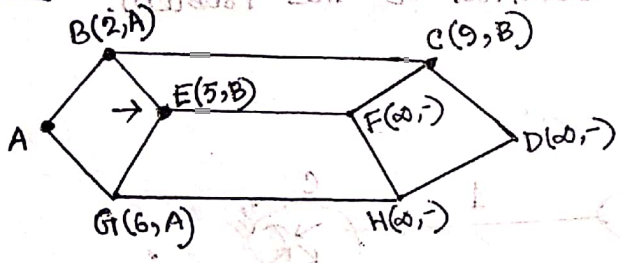
**Step-1:**



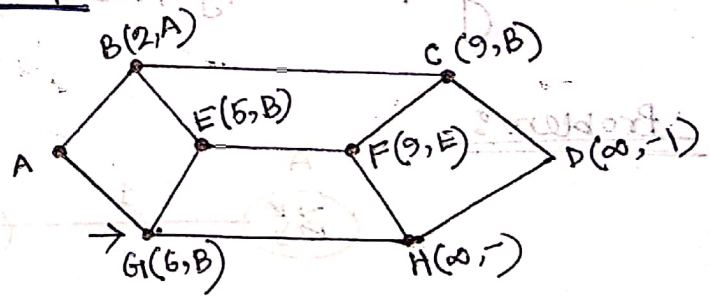

**Step-2:**



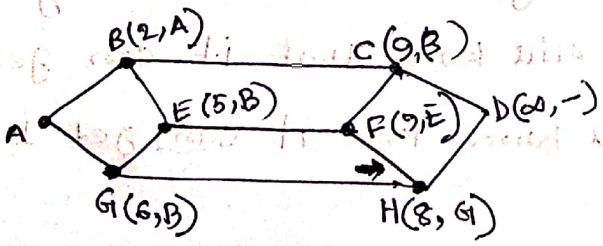

**Step-3:**



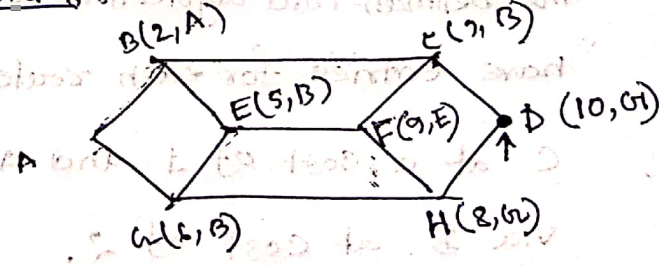

**Step-4:**




**Step-5:**




**Step-6:**




A → B → E → G → H → D

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## 7. Bellman Ford's algorithm:

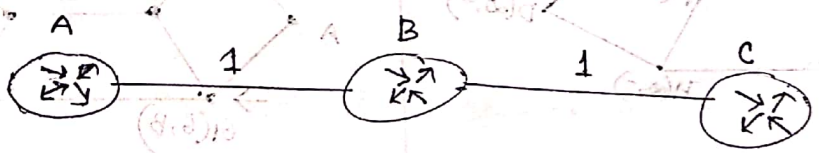
**Algorithm:**

1. The outer loop traverse from 0 : n-1
2. Loop over all edges, Check if the next node distance > current node + edge weight, in this case update the next ~~do~~ node distance to " current node distance + edge weight.

**Example:** (Youtube)

## 8. Counting to infinity Problem and solution of the Problem.

**Problem:**




The Bellman-Ford algorithm will converge for each router, they will have entries for each router. B will know that it can get to C at a cost of 1 and A will know that it can get to C via B at cost of 2.

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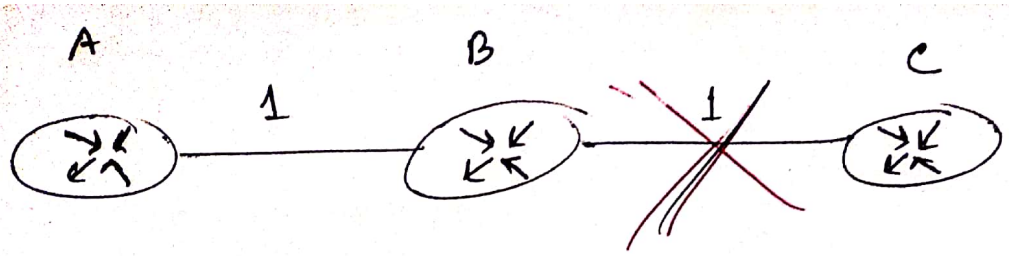


If the link between B & C is disconnected, then B will know that it can no longer get to C via that link and will remove it from the table. Before it can send any updates it's possible that it will recieve an update from A which will be advertising that it can get to C at cost of 2. B can get to A at a cost 1, so it will updated a route to C via A at a cost of 3. A will then recieve updates from B later and update its cost to 4.

They will then go on feeding each other bad information toward infinity which is called as Count to Infinity Problem.

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